

Project ▾
Lab_Thread_Counter D:\JAVA\Lab_Thr
 > .idea
 > out
 > outputs
 TC1_safe.txt
 TC1_unsafe_run1.txt
 TC1_unsafe_run2.txt
 TC1_unsafe_run3.txt
 TC1_unsafe_run4.txt
 TC1_unsafe_run5.txt
 TC2_safe.txt
 TC2_unsafe_run1.txt

TC1_safe.txt x
1 Threads: 1
2 Increments per thread: 1000
3 Mode: Thread-safe
4 Expected count: 1000
5 Static count: 1000
6 Non-static total: 1000
7 Absolute difference: 0
8 Percentage difference: 0.00%

Run ThreadCounterUnsafe x

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=51603" -Dfile.encoding=UTF-8 -Dsun.st  
Threads: 1  
Increments per thread: 1000  
Mode: Thread-safe  
Expected count: 1000  
Static count: 1000  
Non-static total: 1000  
Absolute difference: 0  
Percentage difference: 0.00%  
  
Process finished with exit code 0
```

Lab_Thread_Counter > outputs > TC1_safe.txt

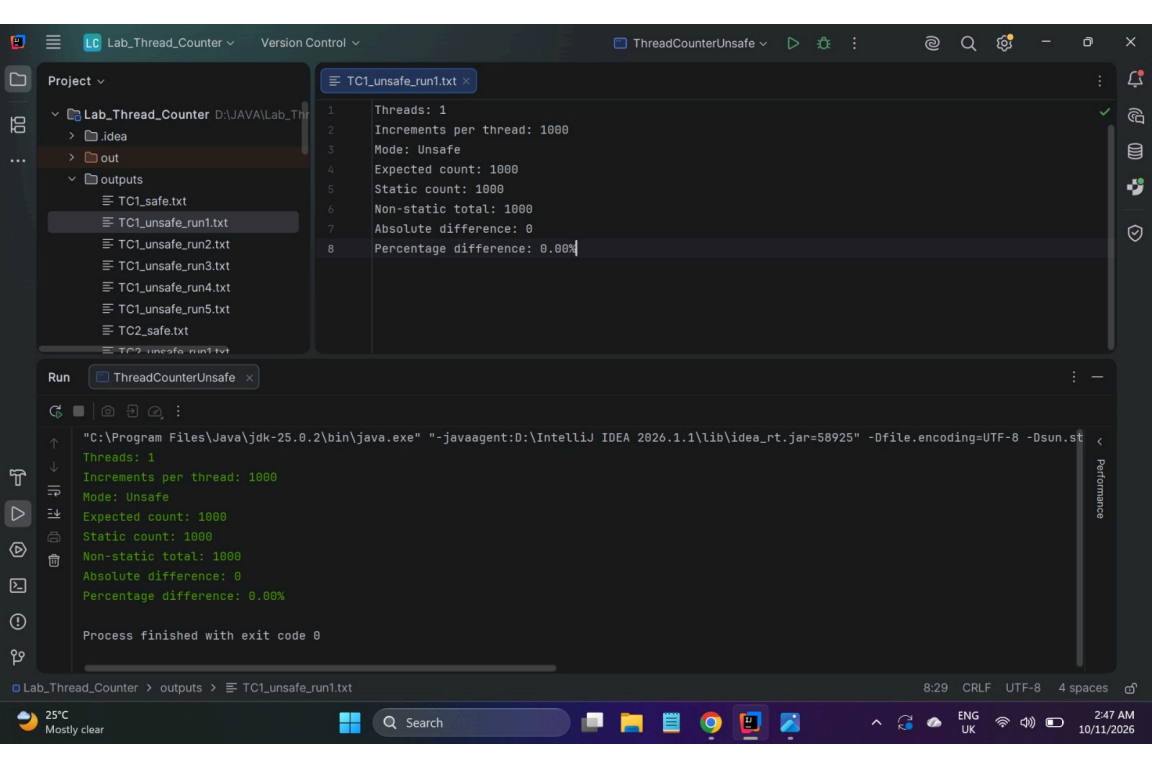
25°C
Mostly clear

Search

8:29 CRLF UTF-8 4 spaces

ENG
UK

2:45 AM
10/11/2026



Project ▾

- Lab_Thread_Counter D:\JAVA\Lab_Thr
 - .idea
 - out
 - outputs
 - TC1_safe.txt
 - TC1_unsafe_run1.txt
 - TC1_unsafe_run2.txt
 - TC1_unsafe_run3.txt
 - TC1_unsafe_run4.txt
 - TC1_unsafe_run5.txt
 - TC2_safe.txt
 - TC2_unsafe_run1.txt

TC1_unsafe_run1.txt ×

```
1 Threads: 1
2 Increments per thread: 1000
3 Mode: Unsafe
4 Expected count: 1000
5 Static count: 1000
6 Non-static total: 1000
7 Absolute difference: 0
8 Percentage difference: 0.00%
```

Run ThreadCounterUnsafe ×

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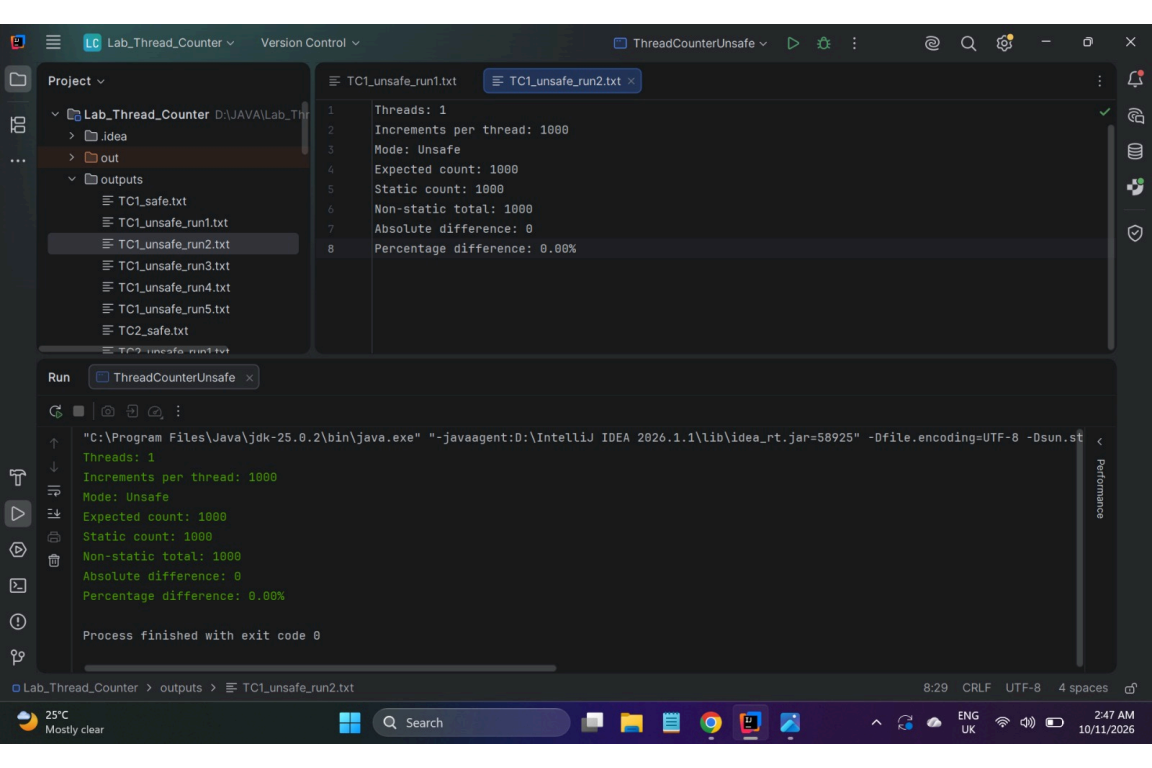
```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=58925" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 1
Increments per thread: 1000
Mode: Unsafe
Expected count: 1000
Static count: 1000
Non-static total: 1000
Absolute difference: 0
Percentage difference: 0.00%

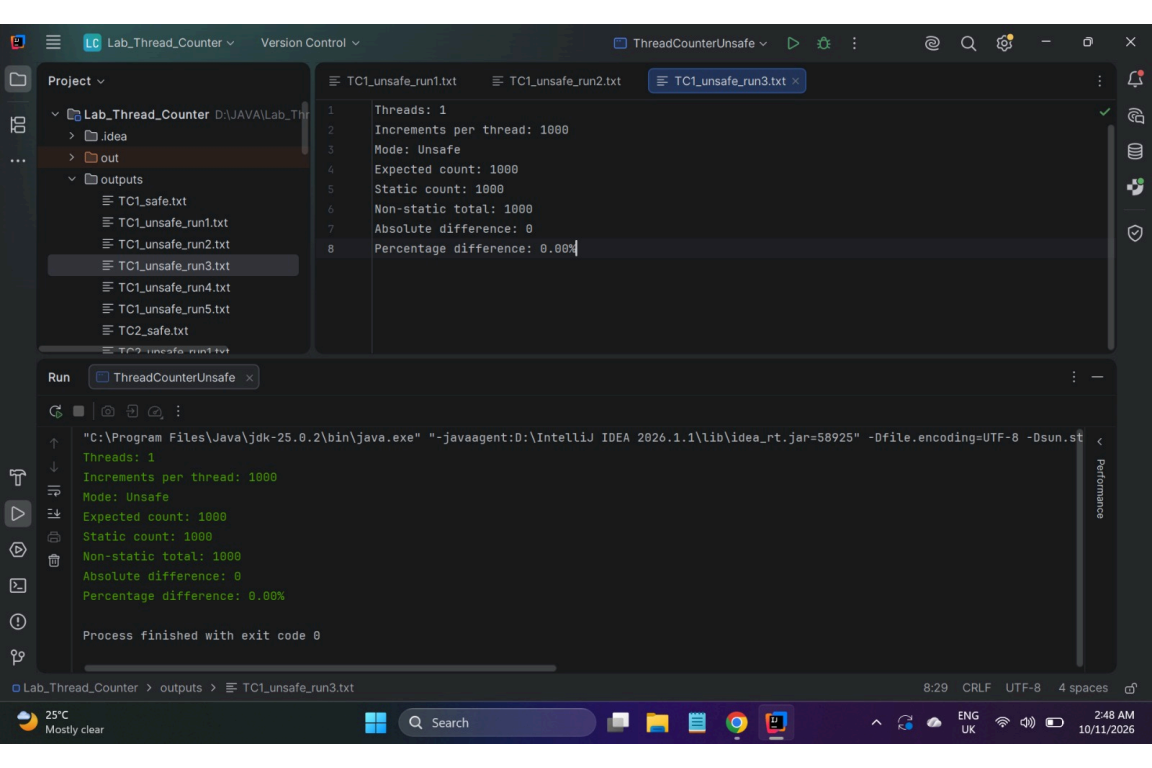
Process finished with exit code 0
```

Performance

Lab_Thread_Counter > outputs > TC1_unsafe_run1.txt

8:29 CRLF UTF-8 4 spaces





Project ▾
+ - ↻ × ⋮ -
Lab_Thread_Counter D:\JAVA\Lab_Thread_Counter
 > .idea
 > out
 > outputs
 TC1_safe.txt
 TC1_unsafe_run1.txt
 TC1_unsafe_run2.txt
 TC1_unsafe_run3.txt
 TC1_unsafe_run4.txt
 TC1_unsafe_run5.txt
 TC2_safe.txt
 TC2_unsafe_run1.txt

TC1_unsafe_run1.txt TC1_unsafe_run2.txt TC1_unsafe_run3.txt TC1_unsafe_run4.txt ×
1 Threads: 1
2 Increments per thread: 1000
3 Mode: Unsafe
4 Expected count: 1000
5 Static count: 1000
6 Non-static total: 1000
7 Absolute difference: 0
8 Percentage difference: 0.00%

Run ThreadCounterUnsafe ×
G [] [] [] [] []
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=58925" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Threads: 1
Increments per thread: 1000
Mode: Unsafe
Expected count: 1000
Static count: 1000
Non-static total: 1000
Absolute difference: 0
Percentage difference: 0.00%

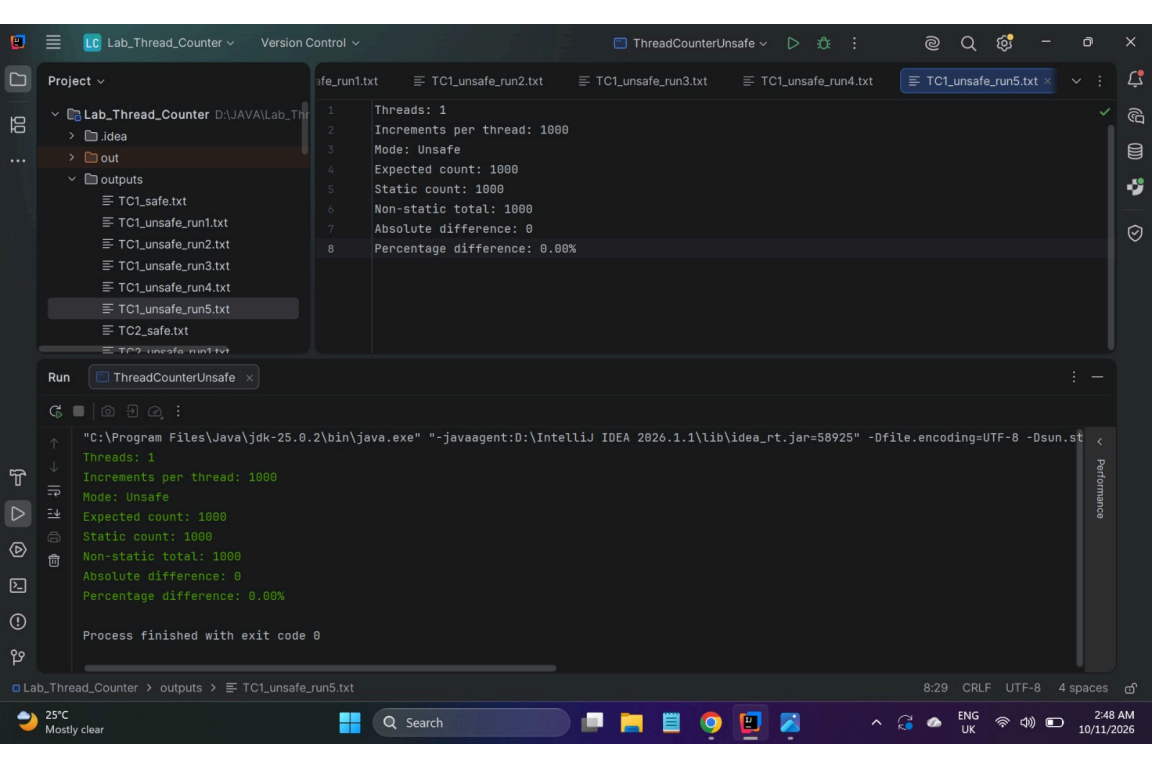
Process finished with exit code 0

Lab_Thread_Counter > outputs > TC1_unsafe_run4.txt 8:29 CRLF UTF-8 4 spaces

25°C Mostly clear

Search

ENG UK 2:48 AM 10/11/2026



Project ▾
Lab_Thread_Counter D:\JAVA\Lab_Thread_Counter
 > .idea
 > out
 > outputs
 TC1_safe.txt
 TC1_unsafe_run1.txt
 TC1_unsafe_run2.txt
 TC1_unsafe_run3.txt
 TC1_unsafe_run4.txt
 TC1_unsafe_run5.txt
 TC2_safe.txt
 TC2_unsafe_run1.txt

TC2_safe.txt x
1 Threads: 2
2 Increments per thread: 10000
3 Mode: Thread-safe
4 Expected count: 20000
5 Static count: 20000
6 Non-static total: 20000
7 Absolute difference: 0
8 Percentage difference: 0.00%

Run ThreadCounterUnsafe x
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=57424" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Threads: 2
Increments per thread: 10000
Mode: Thread-safe
Expected count: 20000
Static count: 20000
Non-static total: 20000
Absolute difference: 0
Percentage difference: 0.00%

Process finished with exit code 0

Lab_Thread_Counter > outputs > TC2_safe.txt

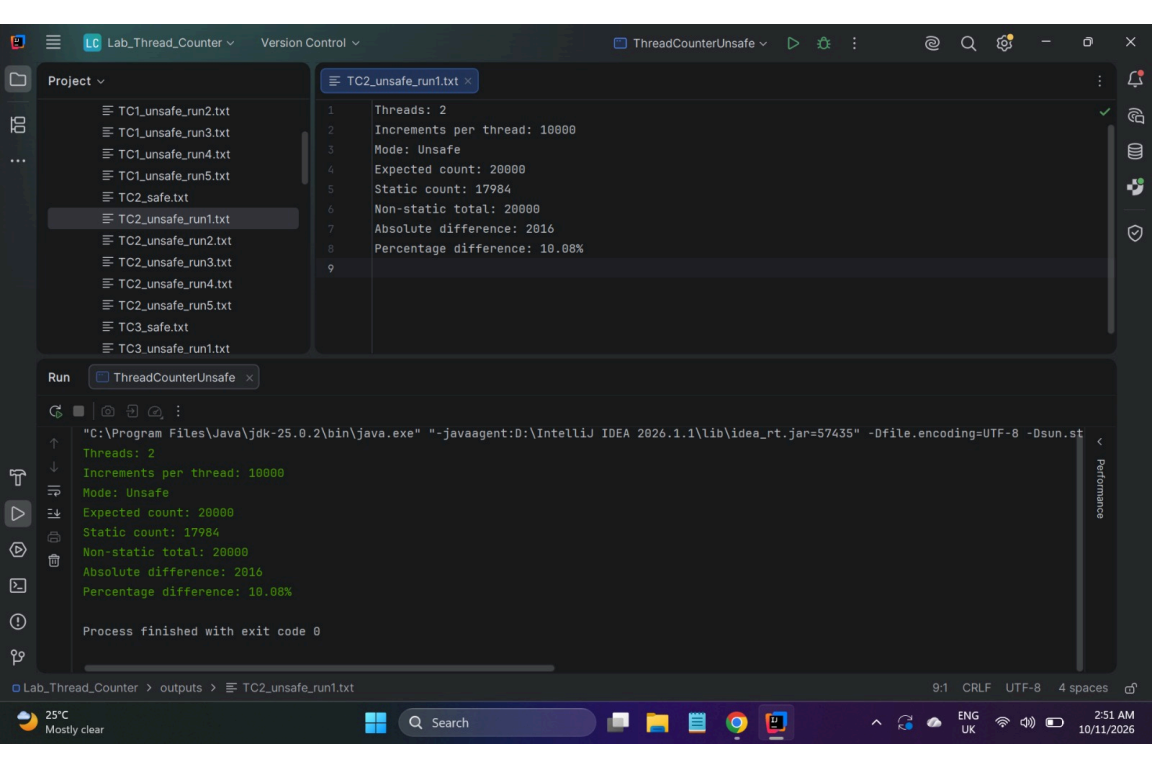
25°C
Mostly clear

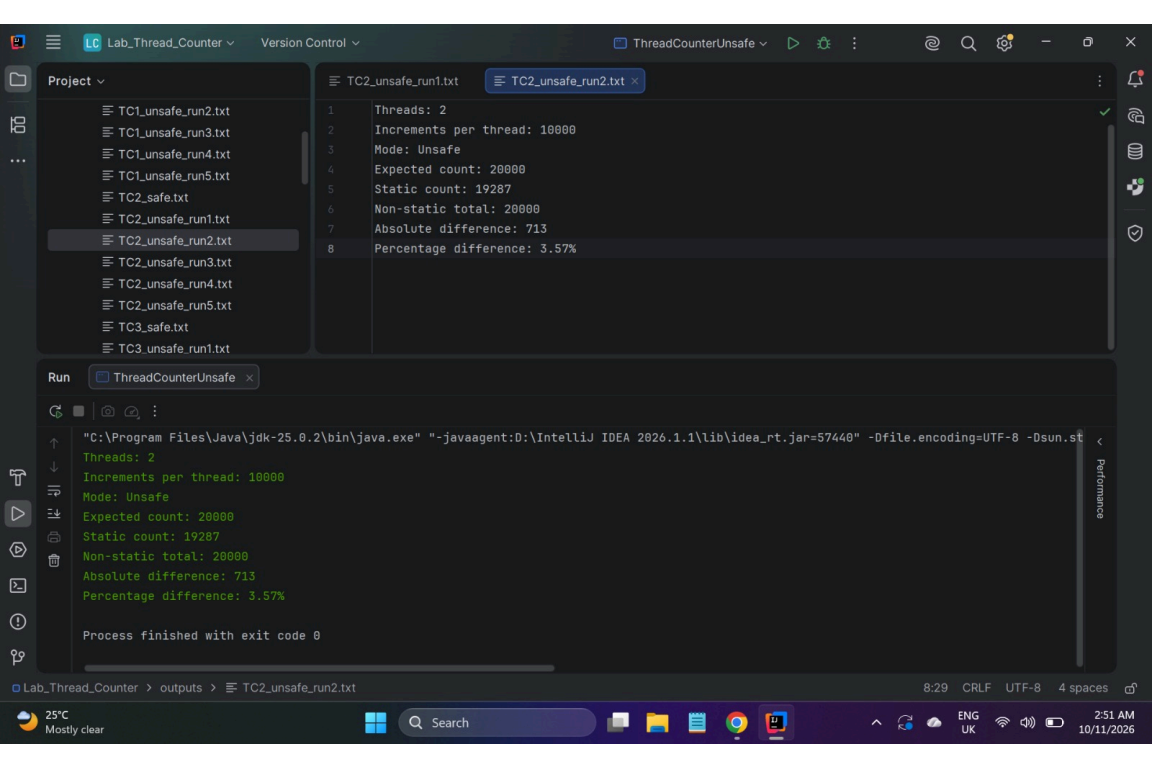
Search

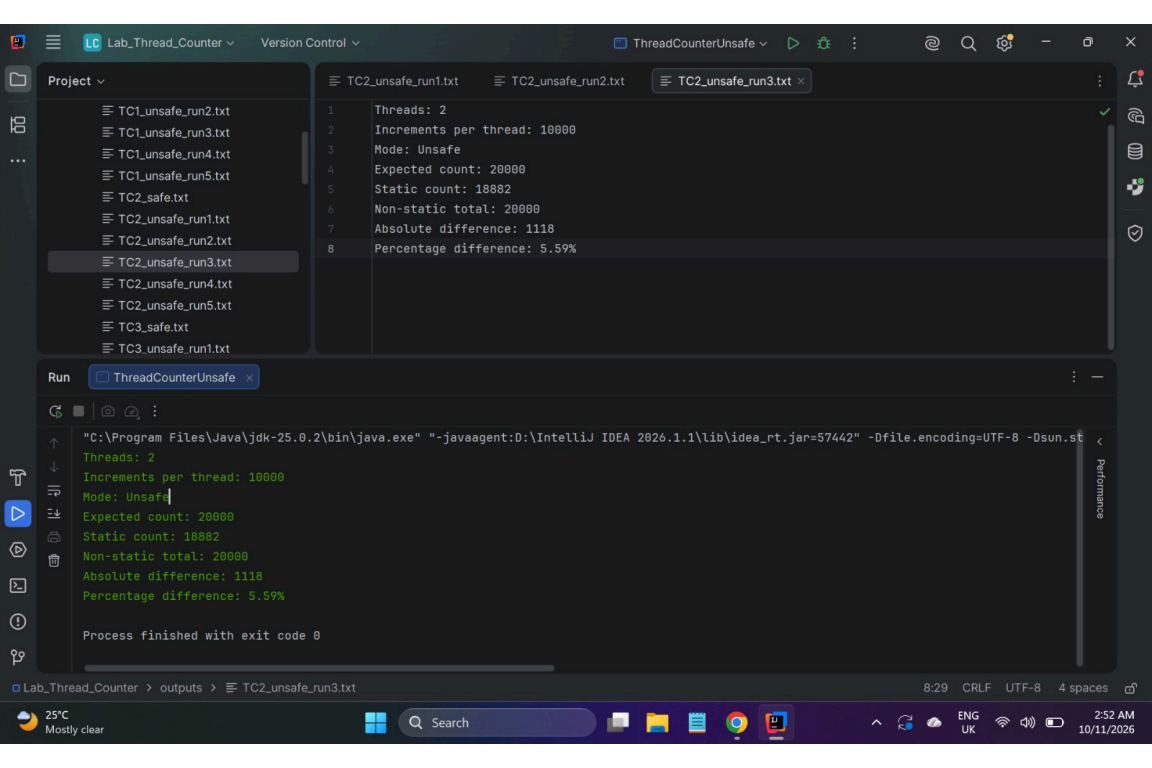
8:29 CRLF UTF-8 4 spaces

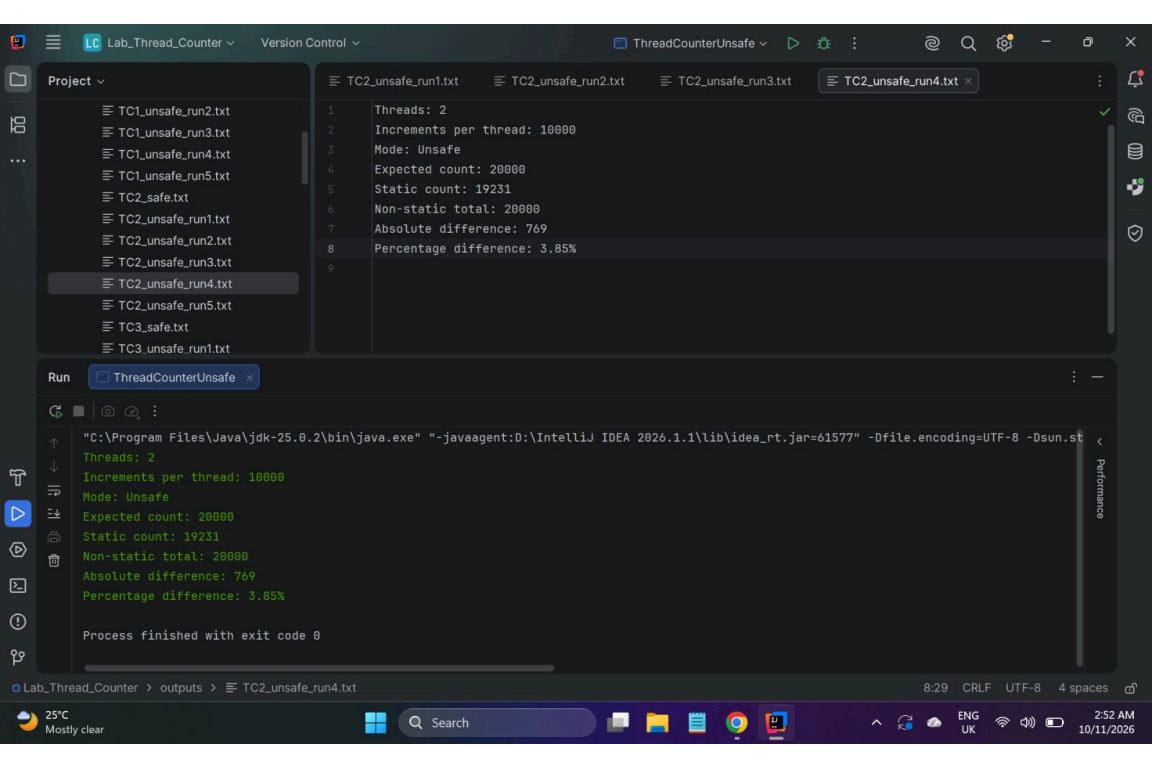
ENG UK

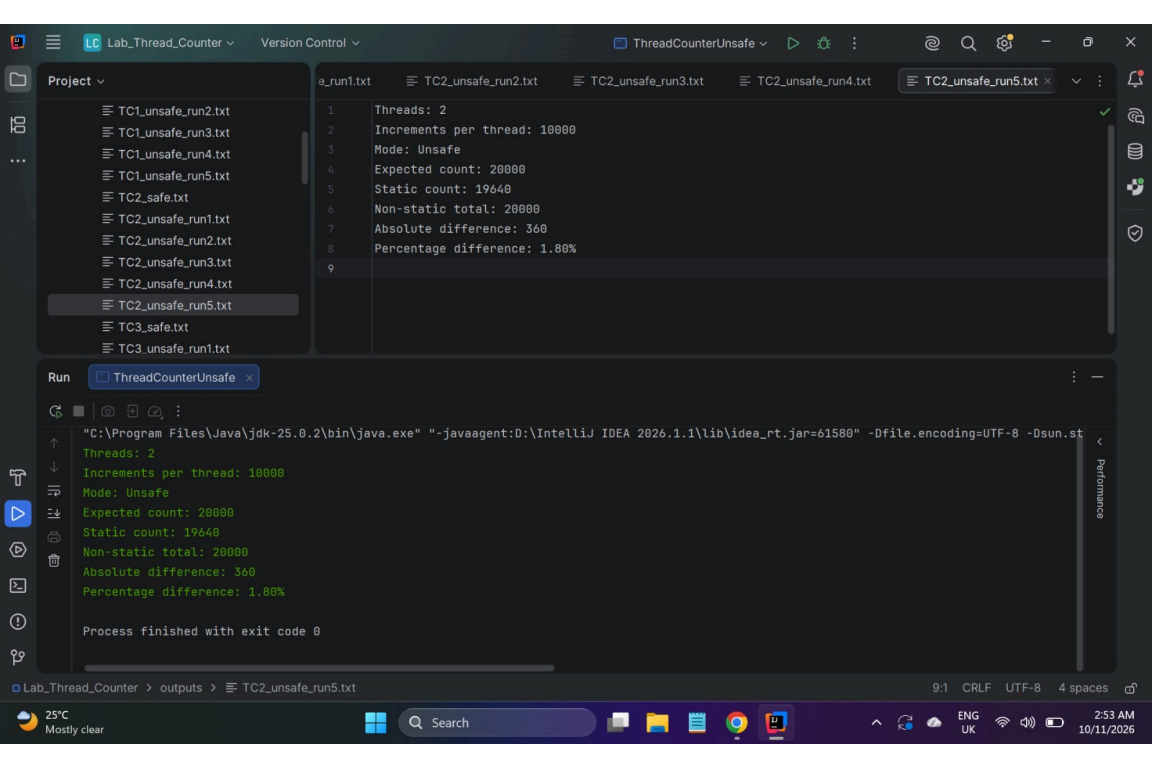
2:50 AM
10/11/2026

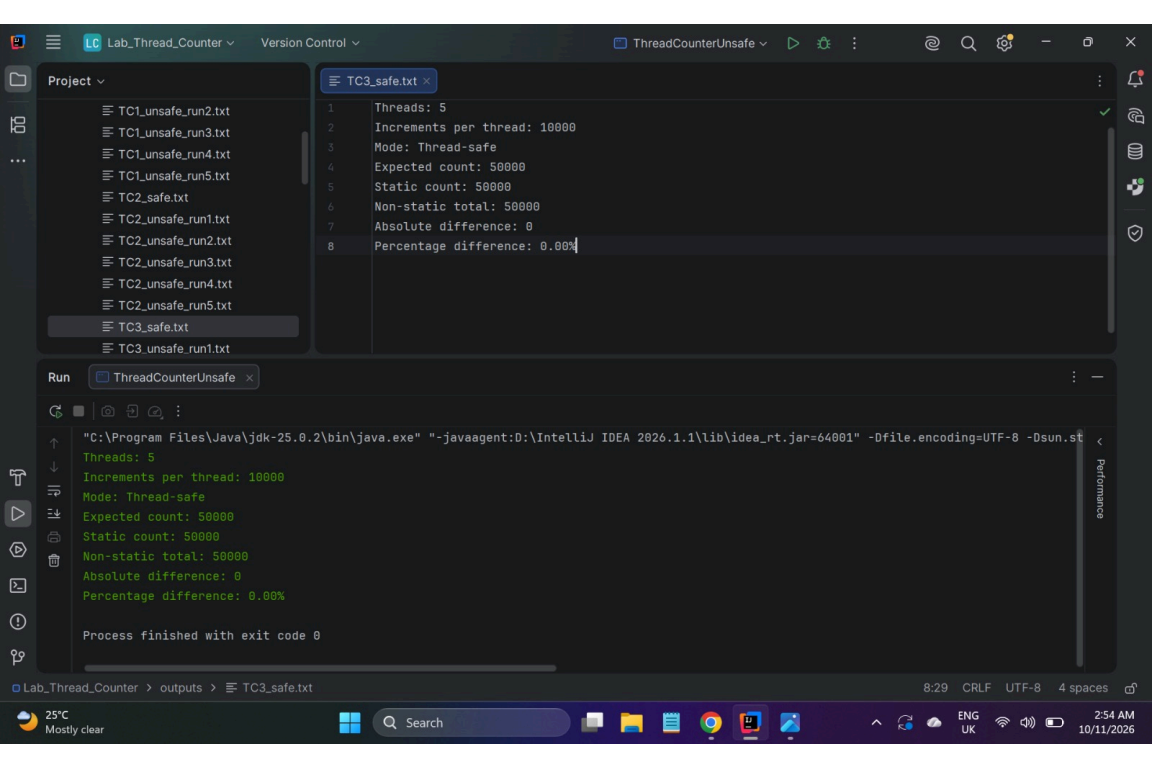


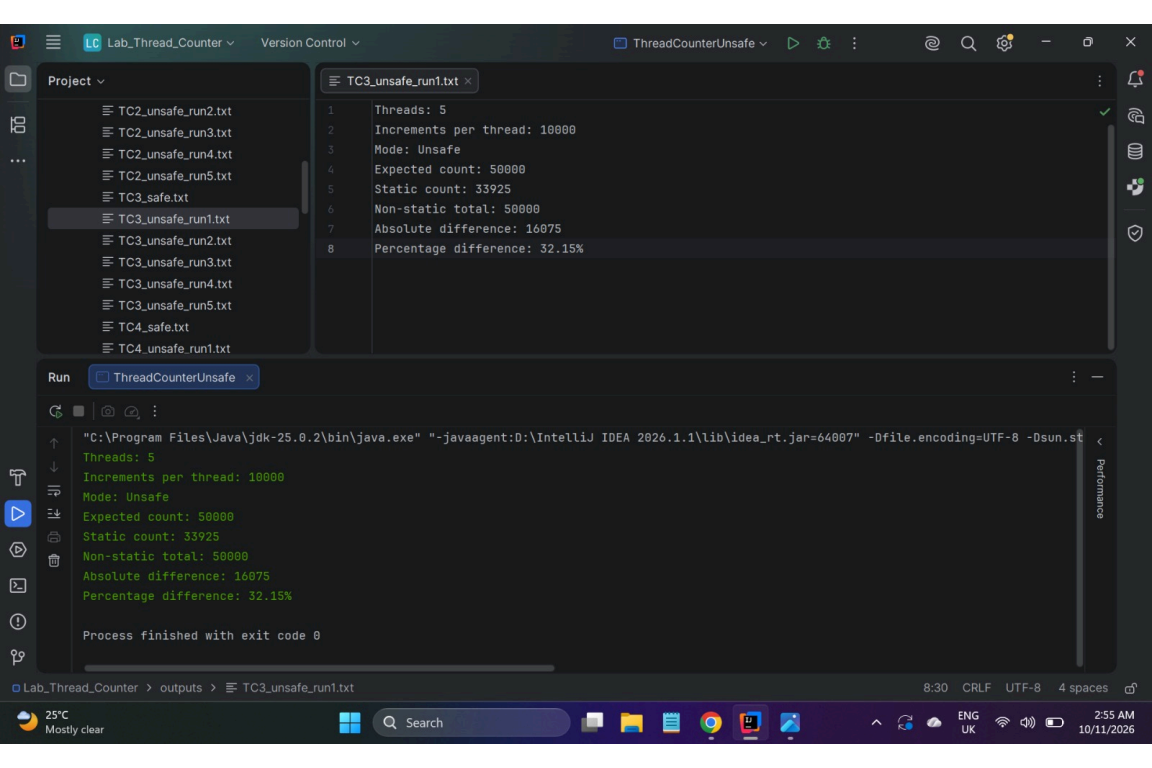












Project ▾

- TC2_unsafe_run2.txt
- TC2_unsafe_run3.txt
- TC2_unsafe_run4.txt
- TC2_unsafe_run5.txt
- TC3_safe.txt
- TC3_unsafe_run1.txt**
- TC3_unsafe_run2.txt
- TC3_unsafe_run3.txt
- TC3_unsafe_run4.txt
- TC3_unsafe_run5.txt
- TC4_safe.txt
- TC4_unsafe_run1.txt

TC3_unsafe_run1.txt ×

```
1 Threads: 5
2 Increments per thread: 10000
3 Mode: Unsafe
4 Expected count: 50000
5 Static count: 33925
6 Non-static total: 50000
7 Absolute difference: 16075
8 Percentage difference: 32.15%
```

Run ThreadCounterUnsafe ×

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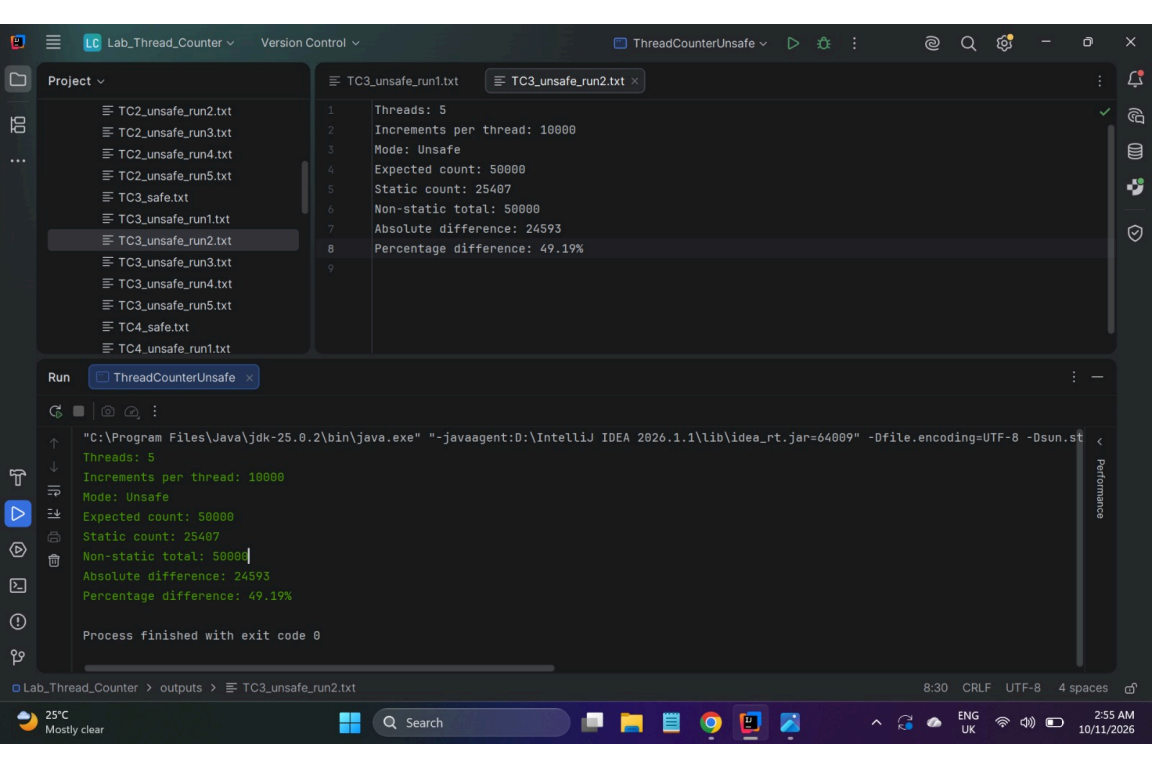
```
↑ "C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64007" -Dfile.encoding=UTF-8 -Dsun.st
↓
Threads: 5
Increments per thread: 10000
Mode: Unsafe
Expected count: 50000
Static count: 33925
Non-static total: 50000
Absolute difference: 16075
Percentage difference: 32.15%

Process finished with exit code 0
```

Performance

Lab_Thread_Counter > outputs > TC3_unsafe_run1.txt

8:30 CRLF UTF-8 4 spaces



Project ▾

- TC2_unsafe_run2.txt
- TC2_unsafe_run3.txt
- TC2_unsafe_run4.txt
- TC2_unsafe_run5.txt
- TC3_safe.txt
- TC3_unsafe_run1.txt
- TC3_unsafe_run2.txt
- TC3_unsafe_run3.txt
- TC3_unsafe_run4.txt
- TC3_unsafe_run5.txt
- TC4_safe.txt
- TC4_unsafe_run1.txt

TC3_unsafe_run1.txt TC3_unsafe_run2.txt TC3_unsafe_run3.txt x

```
1 Threads: 5
2 Increments per thread: 10000
3 Mode: Unsafe
4 Expected count: 50000
5 Static count: 45813
6 Non-static total: 50000
7 Absolute difference: 4187
8 Percentage difference: 8.37%
9
```

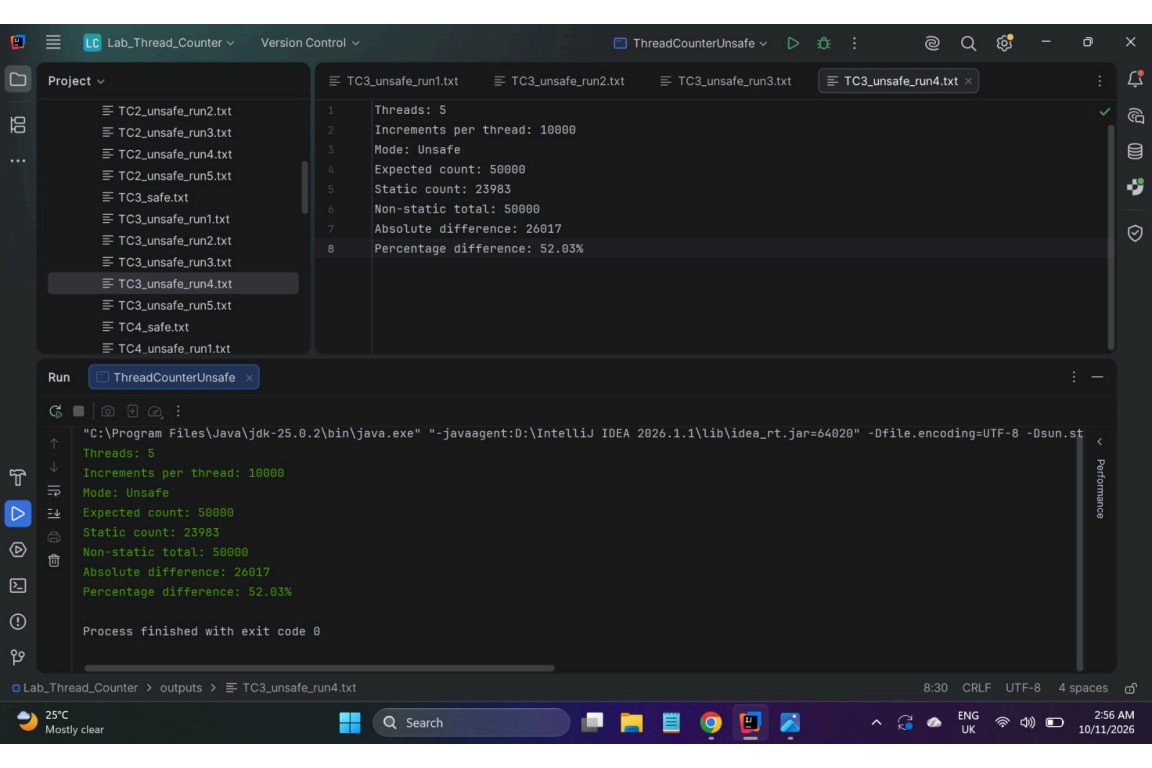
Run ThreadCounterUnsafe x

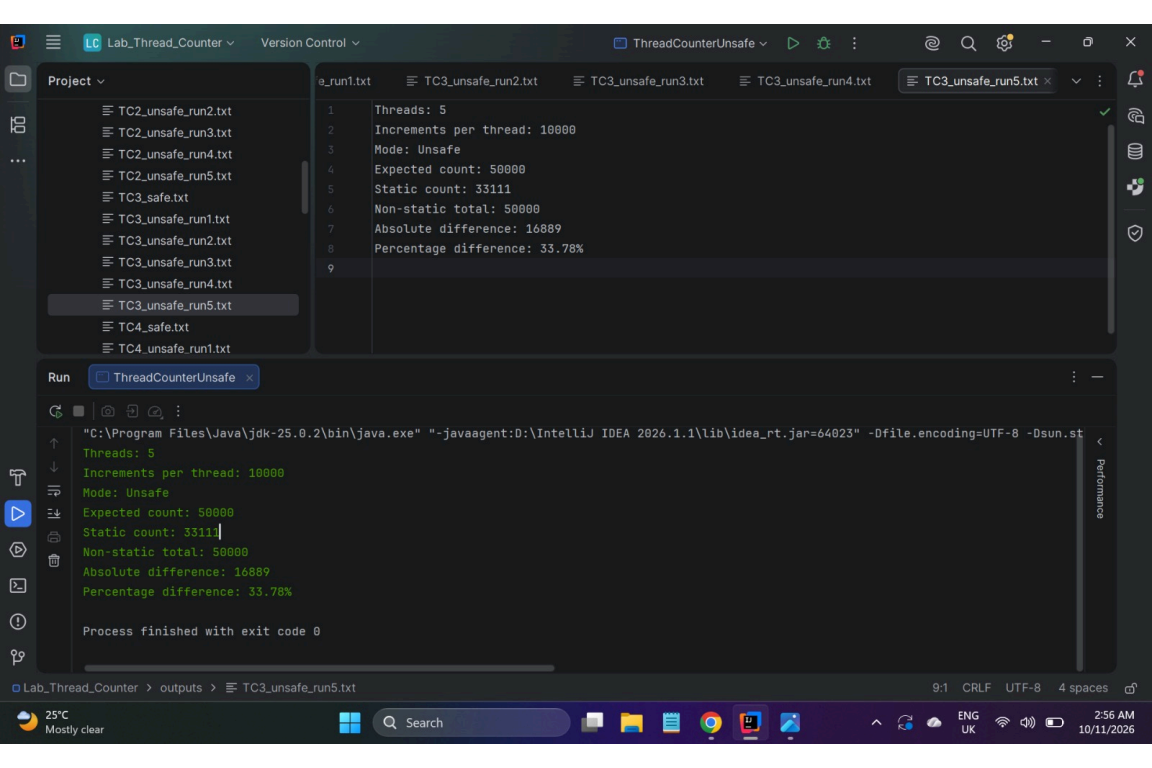
```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64017" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 5
Increments per thread: 10000
Mode: Unsafe
Expected count: 50000
Static count: 45813
Non-static total: 50000
Absolute difference: 4187
Percentage difference: 8.37%

Process finished with exit code 0
```

Lab_Thread_Counter > outputs > TC3_unsafe_run3.txt

25°C Mostly clear 9:1 CRLF UTF-8 4 spaces ENG UK 2:55 AM 10/11/2026





Project ▾

- TC2_unsafe_run2.txt
- TC2_unsafe_run3.txt
- TC2_unsafe_run4.txt
- TC2_unsafe_run5.txt
- TC3_safe.txt
- TC3_unsafe_run1.txt
- TC3_unsafe_run2.txt
- TC3_unsafe_run3.txt
- TC3_unsafe_run4.txt
- TC3_unsafe_run5.txt
- TC4_safe.txt
- TC4_unsafe_run1.txt

e_run1.txt

TC3_unsafe_run2.txt

TC3_unsafe_run3.txt

TC3_unsafe_run4.txt

TC3_unsafe_run5.txt

```
1 Threads: 5
2 Increments per thread: 10000
3 Mode: Unsafe
4 Expected count: 50000
5 Static count: 33111
6 Non-static total: 50000
7 Absolute difference: 16889
8 Percentage difference: 33.78%
9
```

Run

ThreadCounterUnsafe

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```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64023" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 5
Increments per thread: 10000
Mode: Unsafe
Expected count: 50000
Static count: 33111
Non-static total: 50000
Absolute difference: 16889
Percentage difference: 33.78%

Process finished with exit code 0
```

Performance

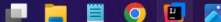
Lab_Thread_Counter > outputs > TC3_unsafe_run5.txt

9:1 CRLF UTF-8 4 spaces

25°C
Mostly clear



Search



ENG
UK



2:56 AM
10/11/2026

Project ▾

- TC2_unsafe_run2.txt
- TC2_unsafe_run3.txt
- TC2_unsafe_run4.txt
- TC2_unsafe_run5.txt
- TC3_safe.txt
- TC3_unsafe_run1.txt
- TC3_unsafe_run2.txt
- TC3_unsafe_run3.txt
- TC3_unsafe_run4.txt
- TC3_unsafe_run5.txt
- TC4_safe.txt
- TC4_unsafe_run1.txt

TC4_safe.txt x

- 1 Threads: 10
- 2 Increments per thread: 50000
- 3 Mode: Thread-safe
- 4 Expected count: 500000
- 5 Static count: 500000
- 6 Non-static total: 500000
- 7 Absolute difference: 0
- 8 Percentage difference: 0.00%

Run ThreadCounterUnsafe x

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64225" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 10
Increments per thread: 50000
Mode: Thread-safe
Expected count: 500000
Static count: 500000
Non-static total: 500000
Absolute difference: 0
Percentage difference: 0.00%

Process finished with exit code 0
```

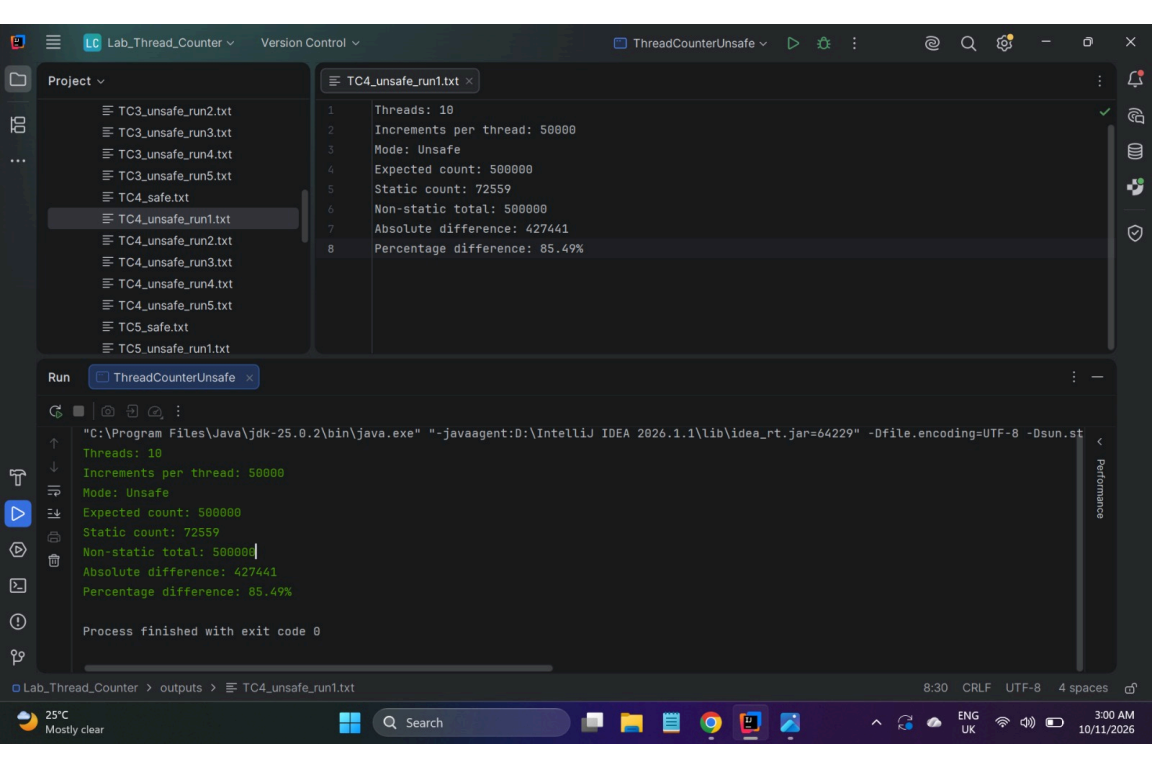
Lab_Thread_Counter > outputs > TC4_safe.txt8:29 CRLF UTF-8 4 spaces

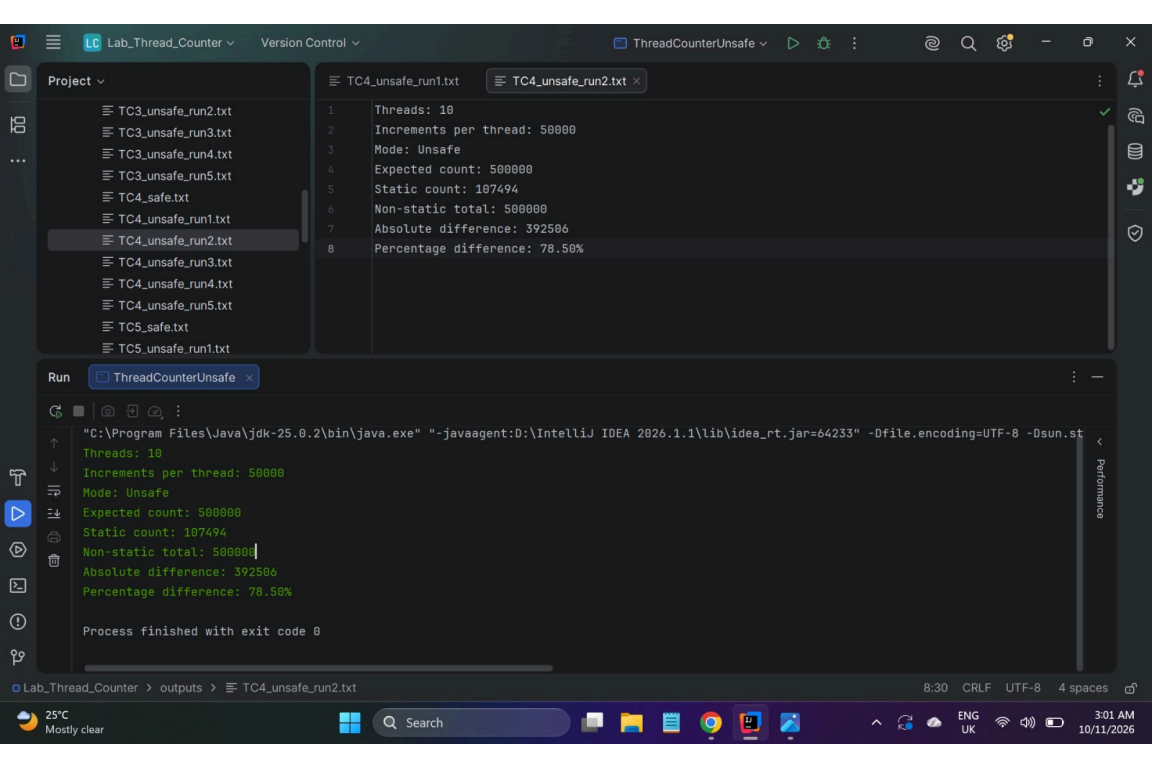
25°C Mostly clear

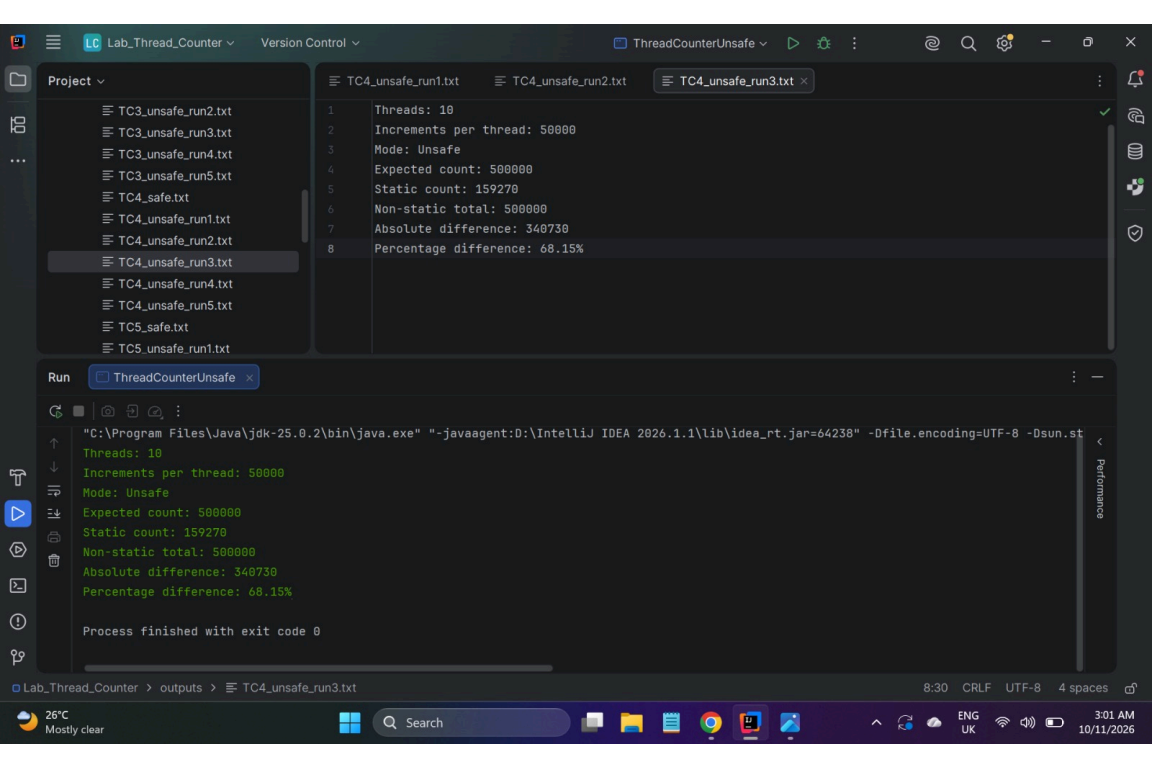
Search

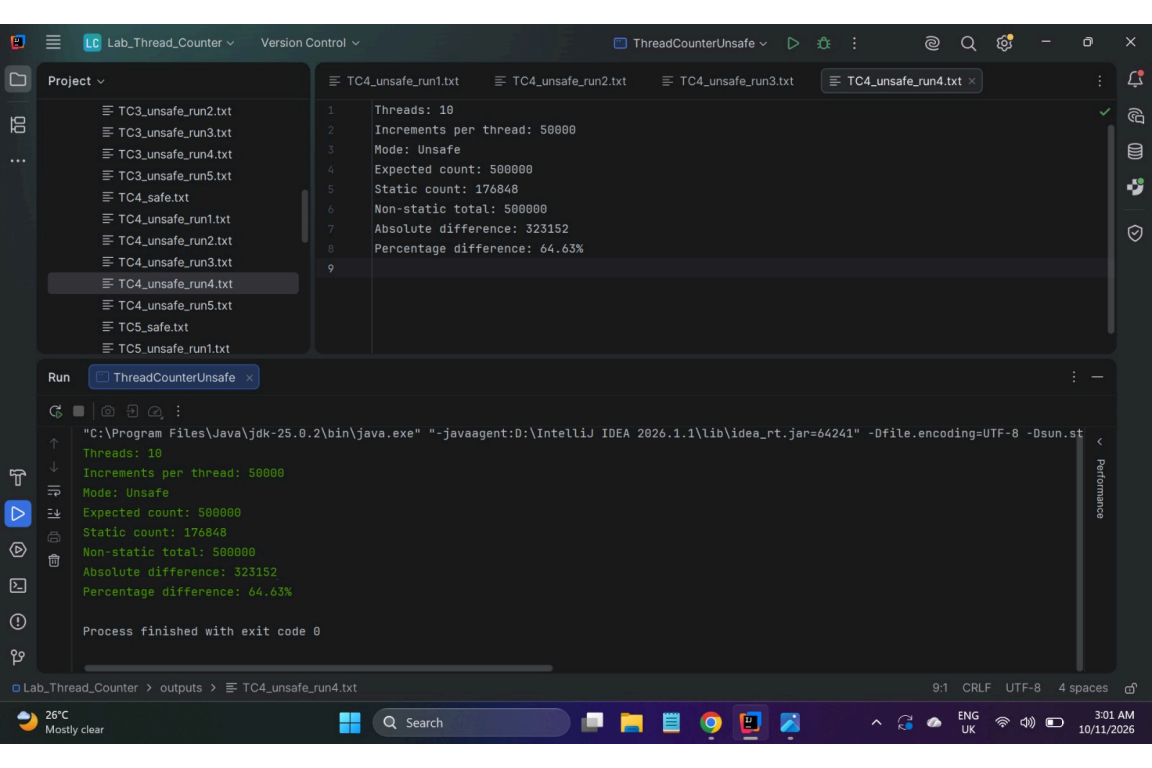
ENG UK

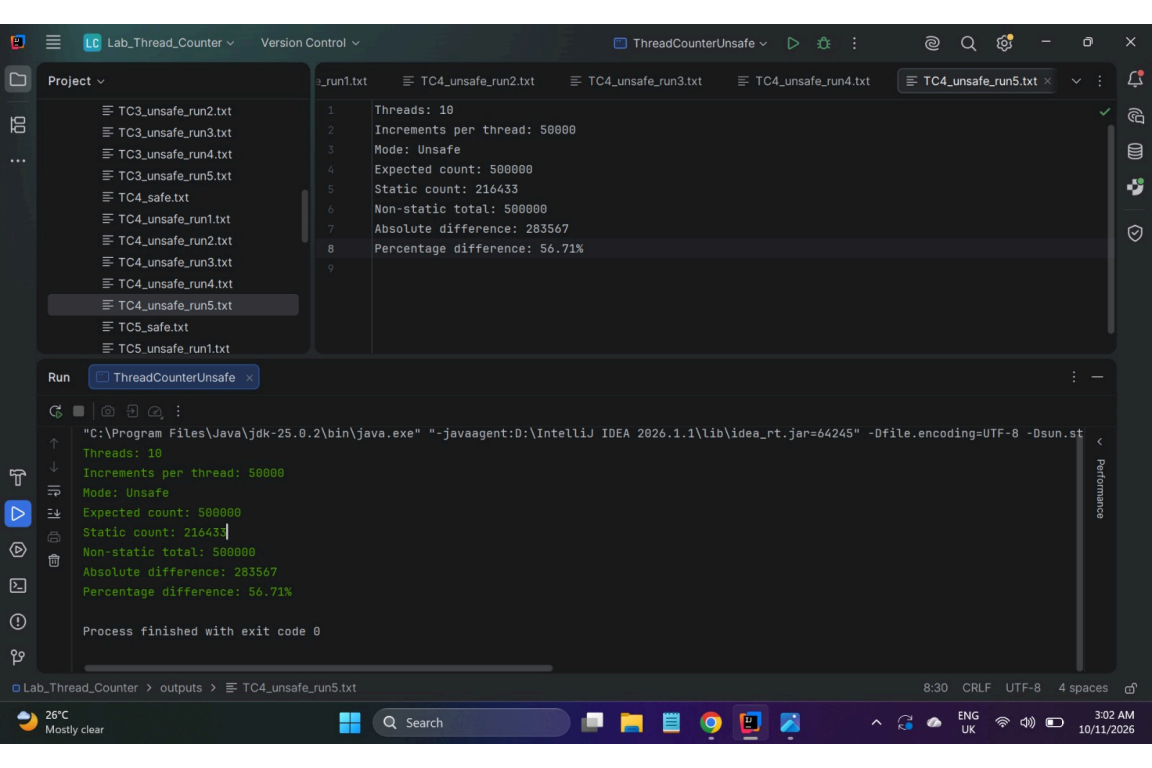
3:00 AM 10/11/2026

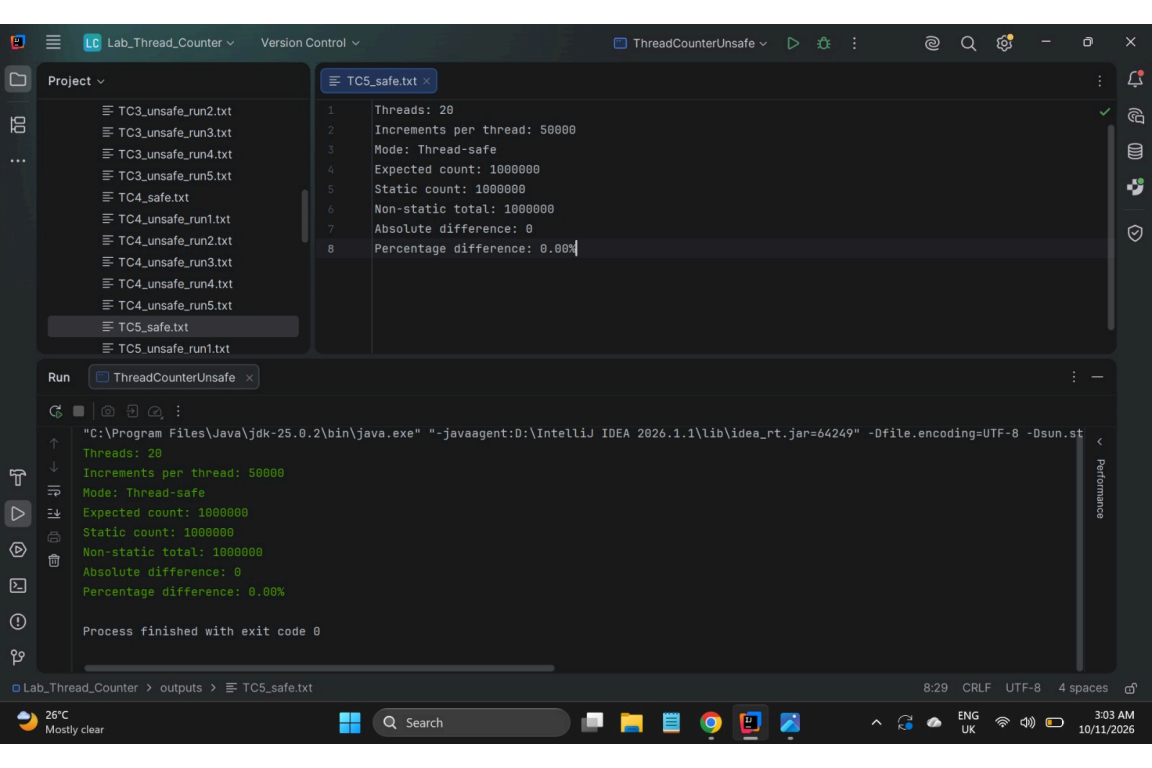


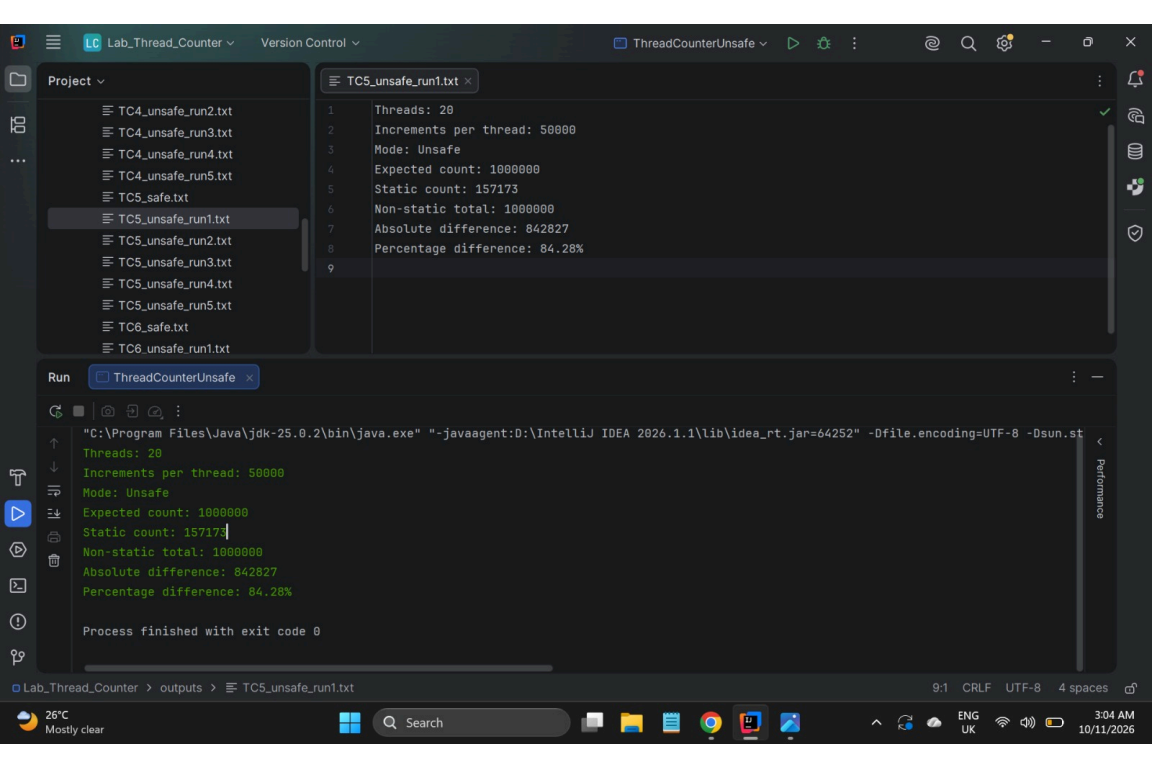


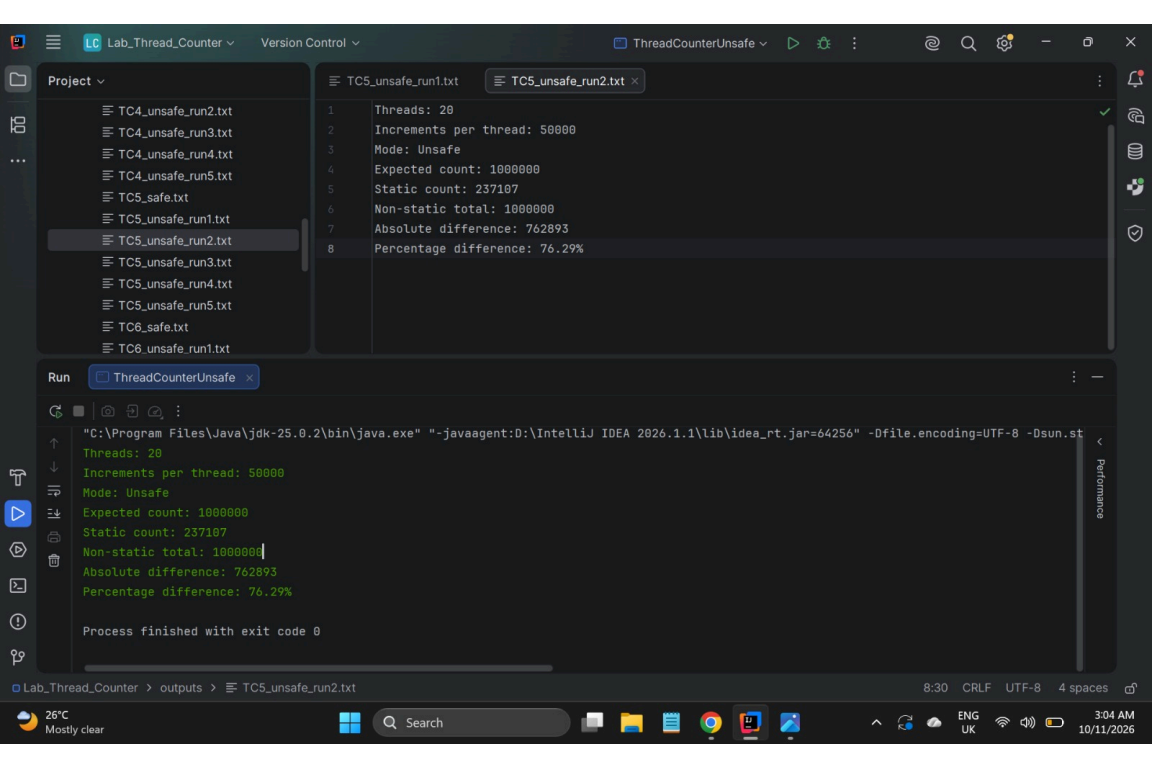


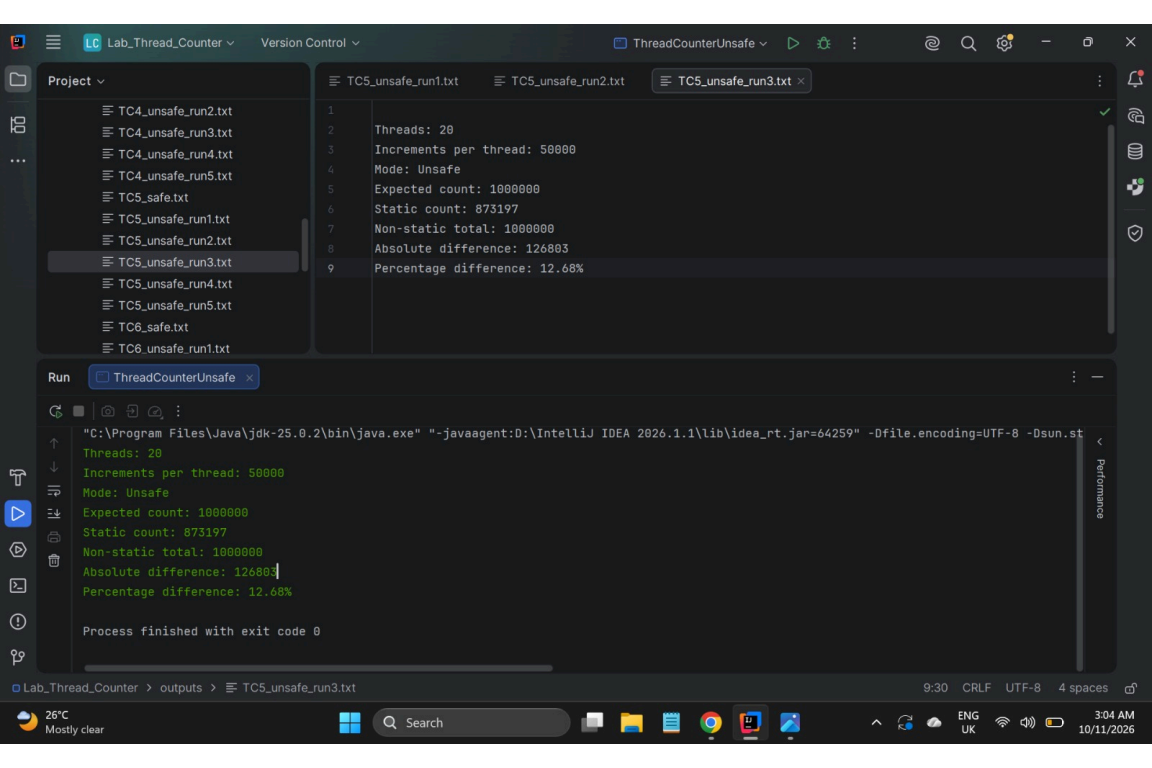


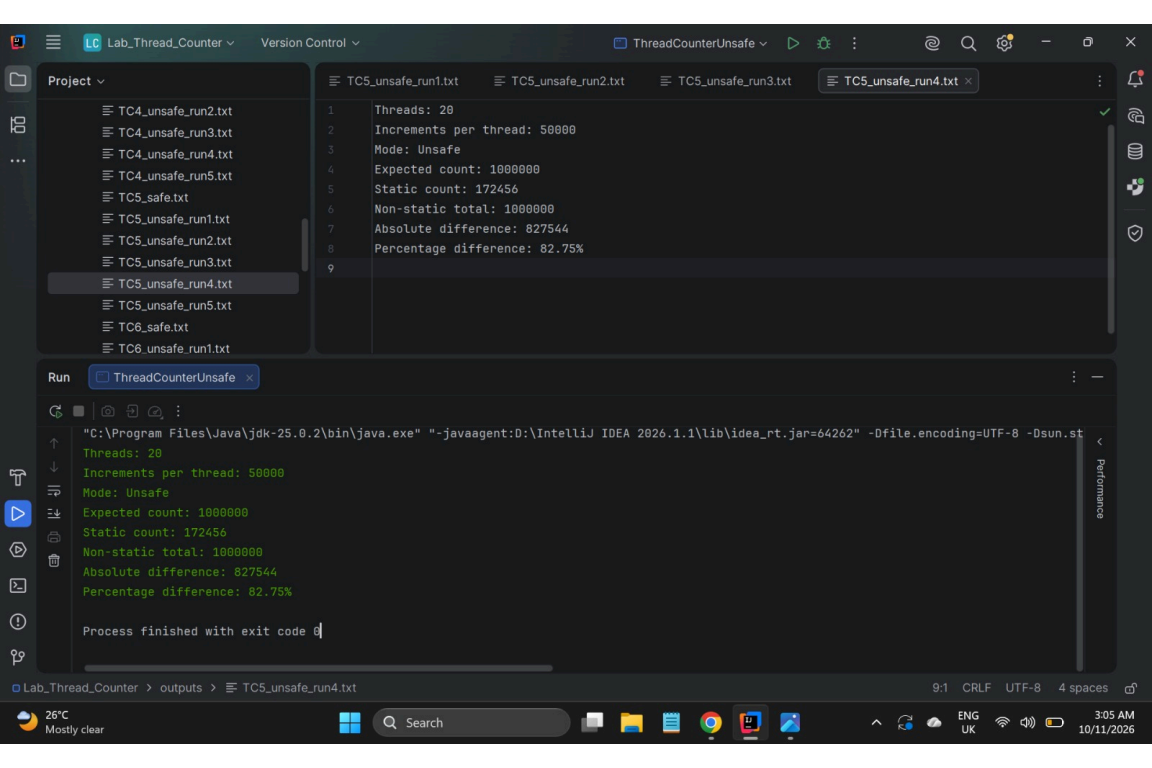


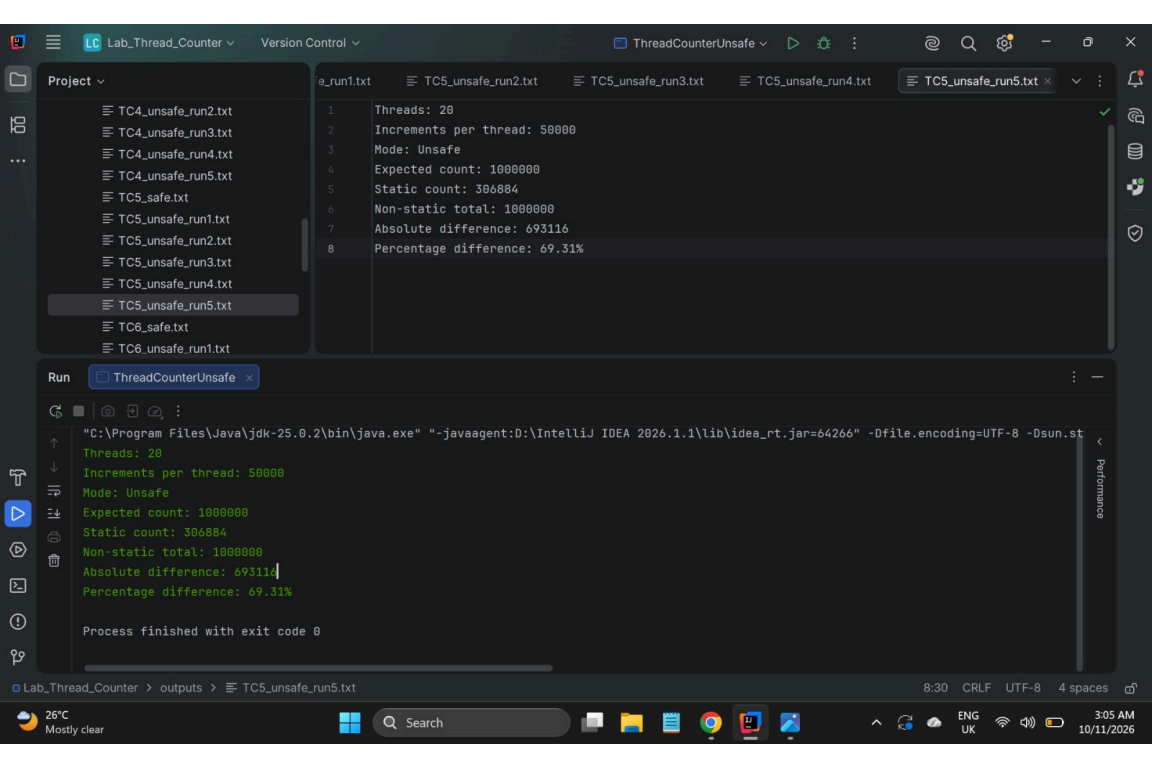


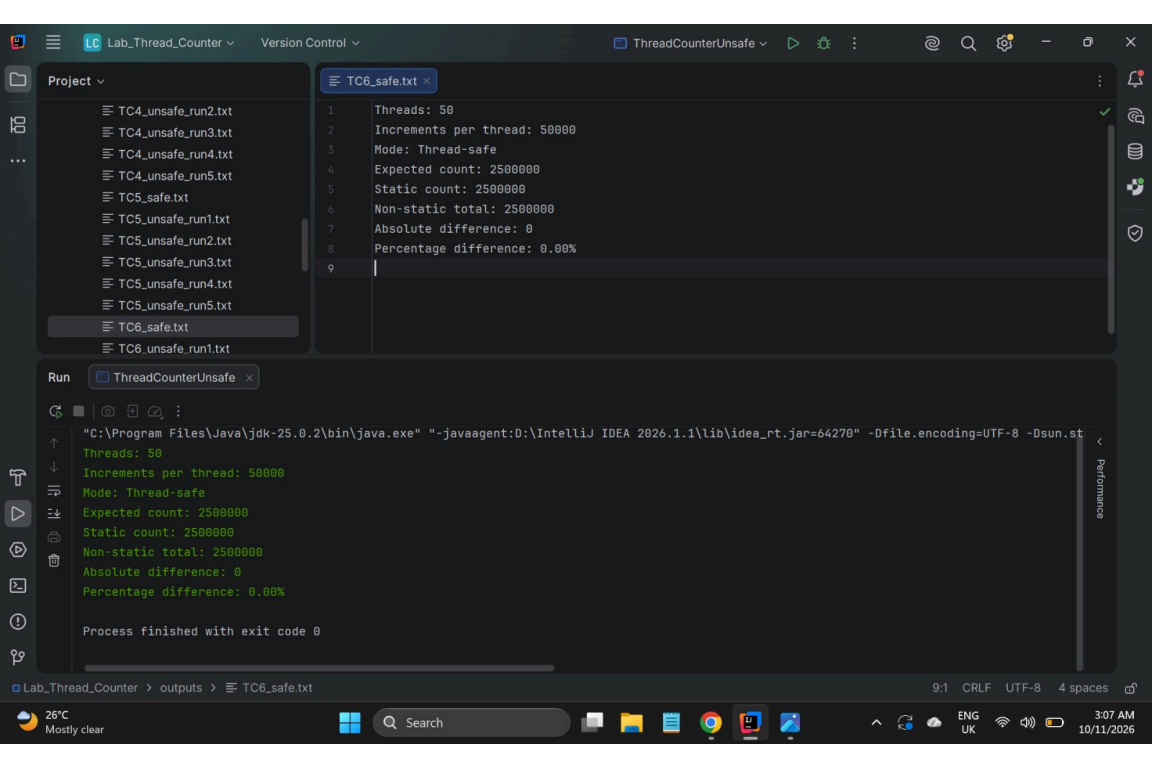


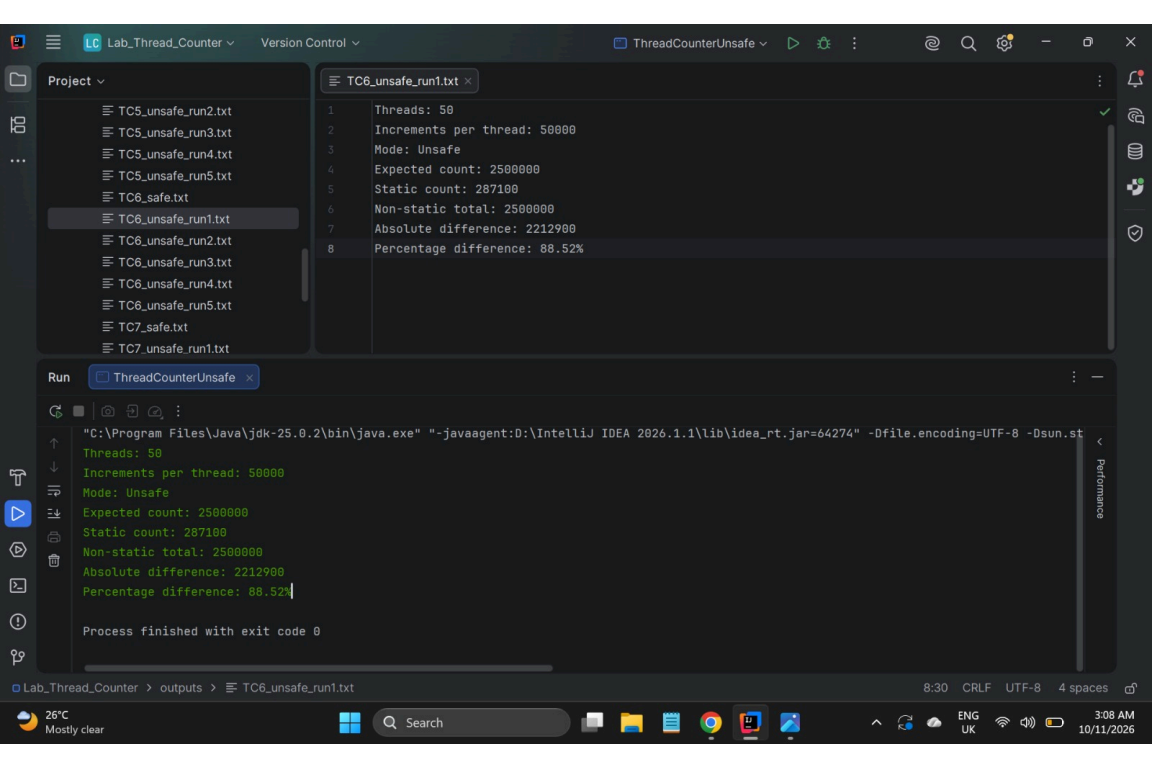


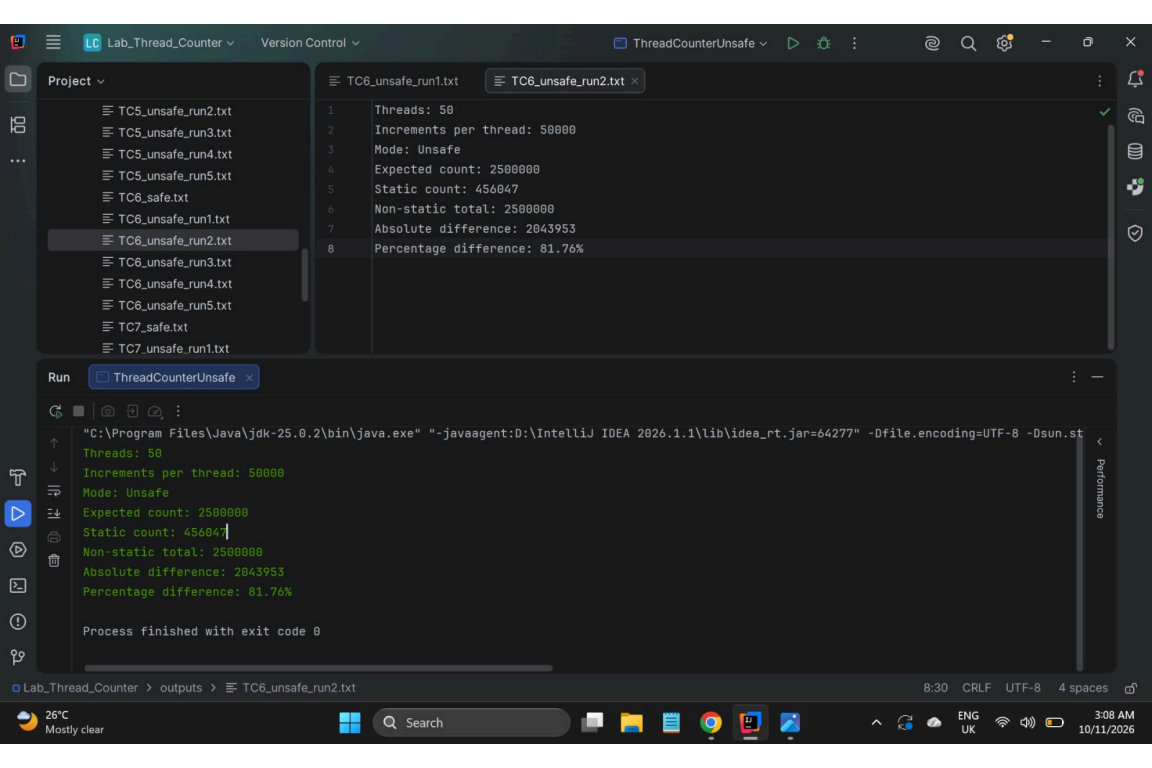


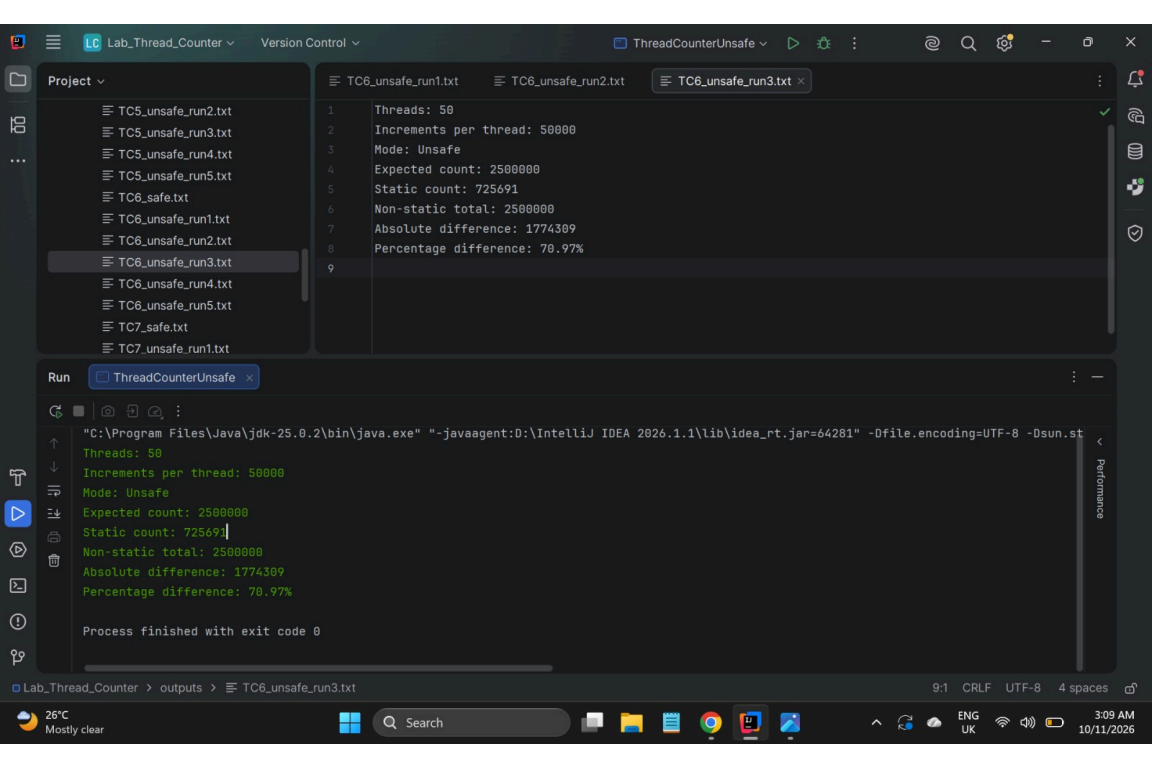












Project ▾

- TC5_unsafe_run2.txt
- TC5_unsafe_run3.txt
- TC5_unsafe_run4.txt
- TC5_unsafe_run5.txt
- TC6_safe.txt
- TC6_unsafe_run1.txt
- TC6_unsafe_run2.txt
- TC6_unsafe_run3.txt
- TC6_unsafe_run4.txt
- TC6_unsafe_run5.txt
- TC7_safe.txt
- TC7_unsafe_run1.txt

TC6_unsafe_run1.txt

TC6_unsafe_run2.txt

TC6_unsafe_run3.txt ×

```
1 Threads: 50
2 Increments per thread: 50000
3 Mode: Unsafe
4 Expected count: 2500000
5 Static count: 725691
6 Non-static total: 2500000
7 Absolute difference: 1774309
8 Percentage difference: 70.97%
9
```

Run

ThreadCounterUnsafe ×

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```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64281" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 50
Increments per thread: 50000
Mode: Unsafe
Expected count: 2500000
Static count: 725691
Non-static total: 2500000
Absolute difference: 1774309
Percentage difference: 70.97%

Process finished with exit code 0
```

Performance

Lab_Thread_Counter > outputs > TC6_unsafe_run3.txt

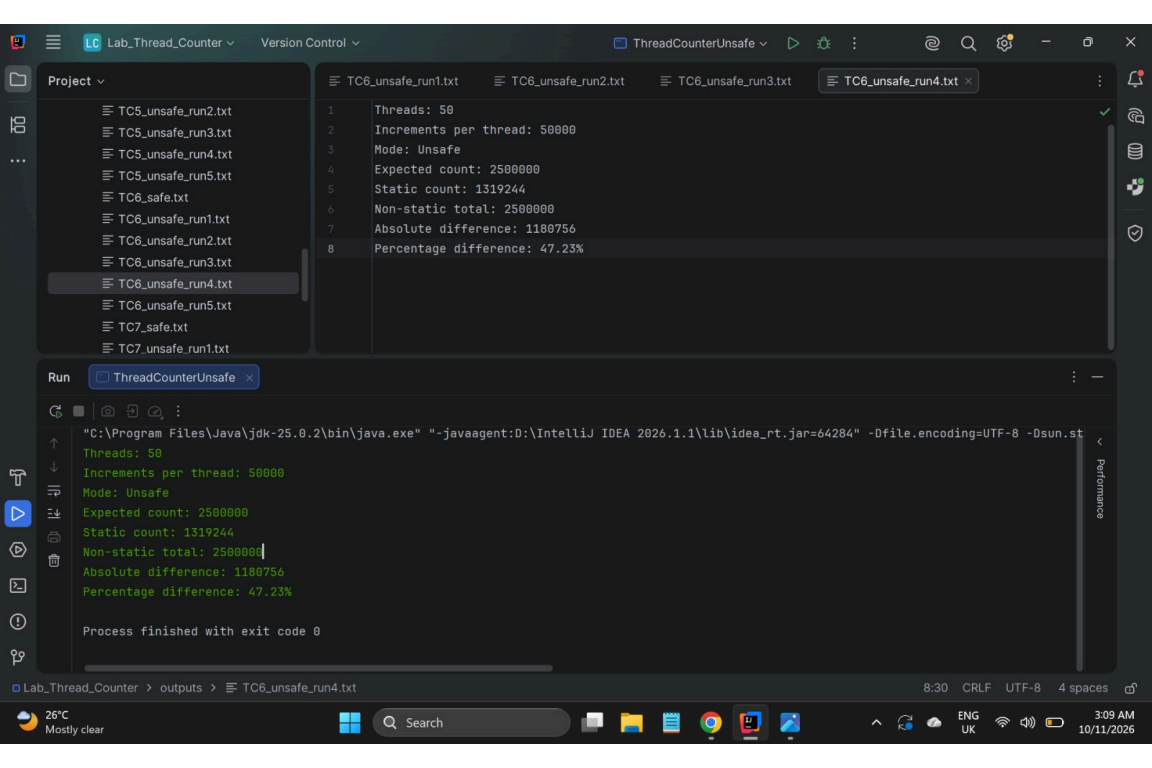
9:1 CRLF UTF-8 4 spaces

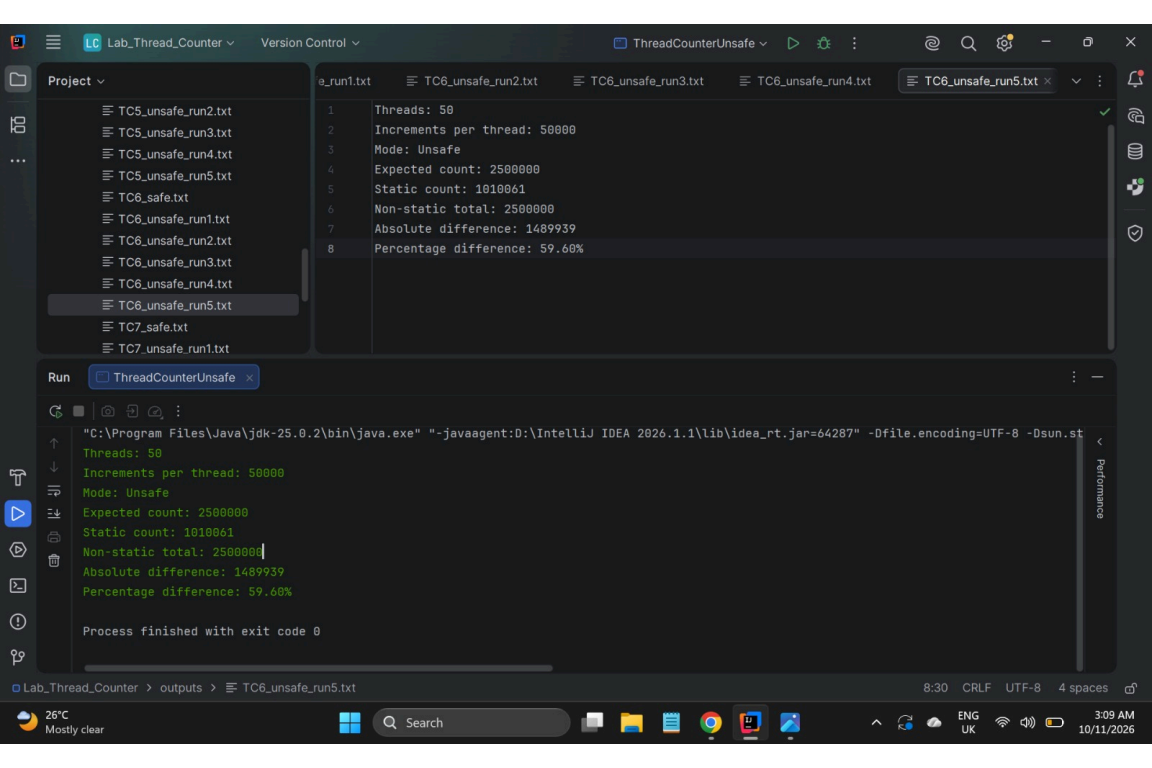
26°C
Mostly clear

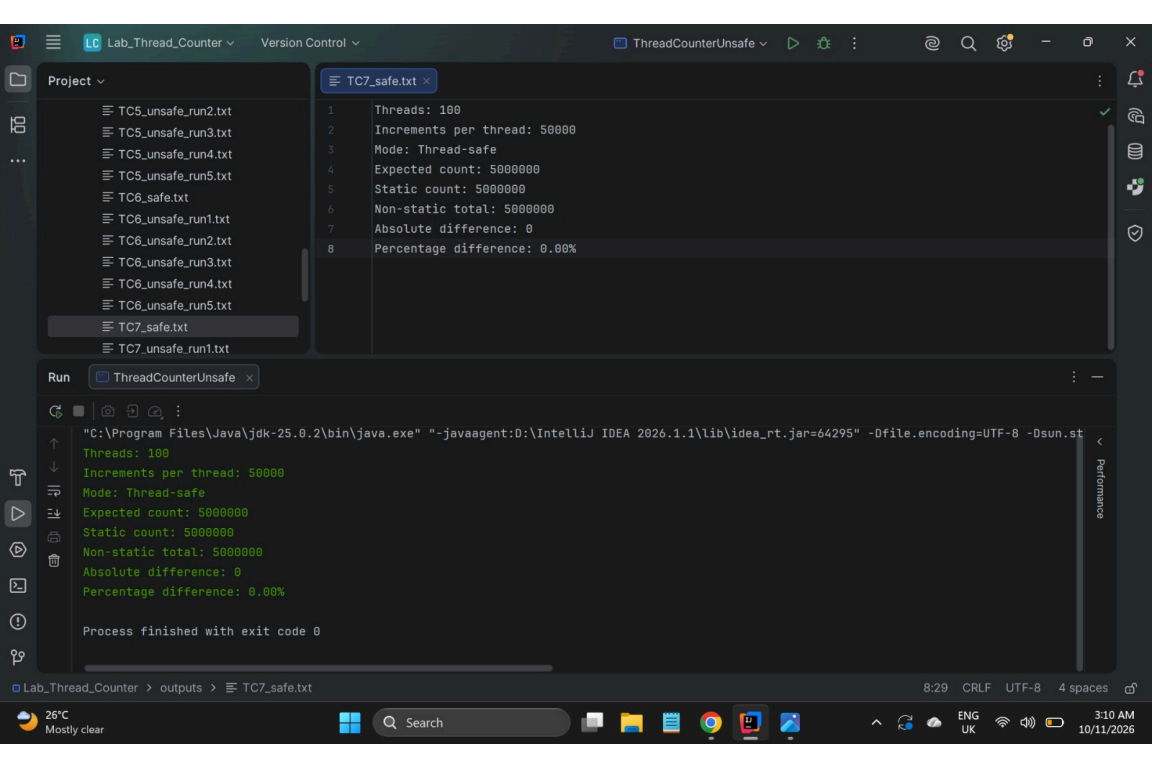
Search

ENG
UK

3:09 AM
10/11/2026







Project ▾

- TC7_unsafe.txt
- TC7_unsafe_run1.txt
- TC7_unsafe_run2.txt
- TC7_unsafe_run3.txt
- TC7_unsafe_run4.txt
- TC7_unsafe_run5.txt
- src 100% classes, 7% lines covered
 - .gitignore
 - Lab_Thread_Counter.iml
- External Libraries
- Scratches and Consoles

TC7_unsafe_run1.txt ×

```
1 Threads: 100
2 Increments per thread: 50000
3 Mode: Unsafe
4 Expected count: 5000000
5 Static count: 3546452
6 Non-static total: 5000000
7 Absolute difference: 1453548
8 Percentage difference: 29.07%
```

Run ThreadCounterUnsafe ×

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64298" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 100
Increments per thread: 50000
Mode: Unsafe
Expected count: 5000000
Static count: 3546452
Non-static total: 5000000
Absolute difference: 1453548
Percentage difference: 29.07%

Process finished with exit code 0
```

Lab_Thread_Counter > outputs > TC7_unsafe_run1.txt

26°C Mostly clear

Search

8:30 CRLF UTF-8 4 spaces

ENG UK

3:11 AM 10/11/2026

Project ▾

- TC7_unsafe.txt
- TC7_unsafe_run1.txt
- TC7_unsafe_run2.txt
- TC7_unsafe_run3.txt
- TC7_unsafe_run4.txt
- TC7_unsafe_run5.txt
- src 100% classes, 7% lines covered
 - .gitignore
 - Lab_Thread_Counter.iml
- External Libraries
- Scratches and Consoles

TC7_unsafe_run1.txt TC7_unsafe_run2.txt ×

```
1 Threads: 100
2 Increments per thread: 50000
3 Mode: Unsafe
4 Expected count: 5000000
5 Static count: 3735151
6 Non-static total: 5000000
7 Absolute difference: 1264849
8 Percentage difference: 25.30%
```

Run ThreadCounterUnsafe ×

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64302" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 100
Increments per thread: 50000
Mode: Unsafe
Expected count: 5000000
Static count: 3735151
Non-static total: 5000000
Absolute difference: 1264849
Percentage difference: 25.30%

Process finished with exit code 0
```

Lab_Thread_Counter > outputs > TC7_unsafe_run2.txt

26°C Mostly clear

Search

8:30 CRLF UTF-8 4 spaces

ENG UK

3:12 AM 10/11/2026

Project ▾

- TC7_unsafe.txt
- TC7_unsafe_run1.txt
- TC7_unsafe_run2.txt
- TC7_unsafe_run3.txt
- TC7_unsafe_run4.txt
- TC7_unsafe_run5.txt
- src 100% classes, 7% lines covered
 - .gitignore
 - Lab_Thread_Counter.iml
- External Libraries
- Scratches and Consoles

TC7_unsafe_run1.txt TC7_unsafe_run2.txt TC7_unsafe_run3.txt ×

```
1 Threads: 100
2 Increments per thread: 50000
3 Mode: Unsafe
4 Expected count: 5000000
5 Static count: 3374905
6 Non-static total: 5000000
7 Absolute difference: 1625095
8 Percentage difference: 32.50%
```

Run ThreadCounterUnsafe ×

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64305" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 100
Increments per thread: 50000
Mode: Unsafe
Expected count: 5000000
Static count: 3374905
Non-static total: 5000000
Absolute difference: 1625095
Percentage difference: 32.50%

Process finished with exit code 0
```

Lab_Thread_Counter > outputs > TC7_unsafe_run3.txt

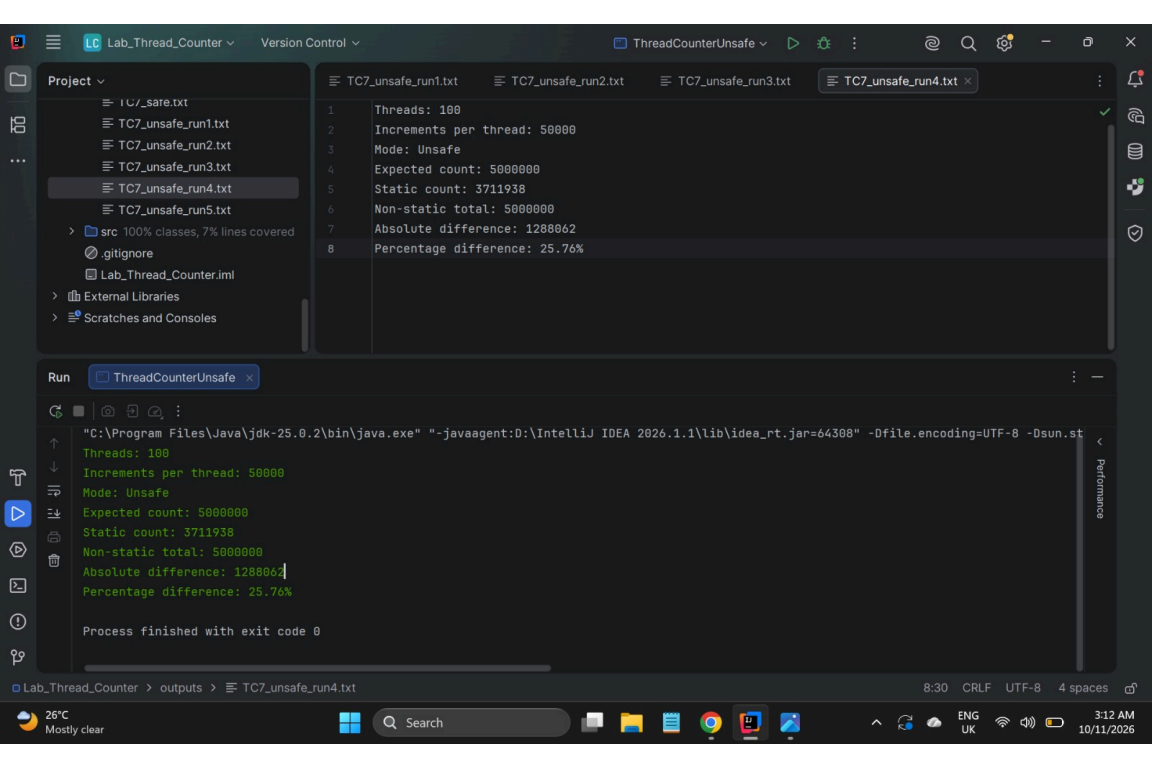
26°C Mostly clear

Search

8:30 CRLF UTF-8 4 spaces

ENG UK

3:12 AM 10/11/2026



Project ▾

- TC7_unsafe.txt
- TC7_unsafe_run1.txt
- TC7_unsafe_run2.txt
- TC7_unsafe_run3.txt
- TC7_unsafe_run4.txt
- TC7_unsafe_run5.txt
- > src 100% classes, 7% lines covered
 - .gitignore
 - Lab_Thread_Counter.iml
- > External Libraries
- > Scratches and Consoles

TC7_unsafe_run1.txt TC7_unsafe_run2.txt TC7_unsafe_run3.txt TC7_unsafe_run4.txt x

```
1 Threads: 100
2 Increments per thread: 50000
3 Mode: Unsafe
4 Expected count: 5000000
5 Static count: 3711938
6 Non-static total: 5000000
7 Absolute difference: 1288062
8 Percentage difference: 25.76%
```

Run ThreadCounterUnsafe x

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```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=64308" -Dfile.encoding=UTF-8 -Dsun.st
Threads: 100
Increments per thread: 50000
Mode: Unsafe
Expected count: 5000000
Static count: 3711938
Non-static total: 5000000
Absolute difference: 1288062
Percentage difference: 25.76%

Process finished with exit code 0
```

Performance

Lab_Thread_Counter > outputs > TC7_unsafe_run4.txt

8:30 CRLF UTF-8 4 spaces

